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SINGLE INSERTION DELIVERY SYSTEM FOR TREATING EMBOLISM AND ASSOCIATED SYSTEMS AND METHODS

Abstract

Systems and methods for the intravascular treatment of clot material within a blood vessel of a human patient are disclosed herein. A method in accordance with embodiments of the present technology can include, for example, engaging an interventional device of a catheter system with clot material in a blood vessel and withdrawing the interventional device and the portion of the clot material through a guide catheter. In some embodiments, the catheter system can include an attachment/valve member coupled to a proximal portion of the guide catheter, and the method can include unsealing the attachment/valve member to facilitate withdrawing the interventional device through the attachment/valve member without significant retention of clot material within the attachment/valve member. The method can further include resealing and aspirating the guide catheter before advancing another interventional device to the clot material to again engage and remove clot material from the blood vessel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/497,249, filed on Oct. 30, 2023, and titled “SINGLE INSERTION DELIVERY SYSTEM FOR TREATING EMBOLISM AND ASSOCIATED SYSTEMS AND METHODS,” which is a continuation of U.S. patent application Ser. No. 17/865,307, filed on Jul. 14, 2022, and titled “SINGLE INSERTION DELIVERY SYSTEM FOR TREATING EMBOLISM AND ASSOCIATED SYSTEMS AND METHODS,” now issued as U.S. Pat. No. 11,849,963, which is a continuation of U.S. patent application Ser. No. 17/498,642, filed on Oct. 11, 2021, and titled “SINGLE INSERTION DELIVERY SYSTEM FOR TREATING EMBOLISM AND ASSOCIATED SYSTEMS AND METHODS,” now issued as U.S. Pat. No. 12,016,580, which is a continuation of U.S. patent application Ser. No. 16/258,344, filed on Jan. 25, 2019, and titled “SINGLE INSERTION DELIVERY SYSTEM FOR TREATING EMBOLISM AND ASSOCIATED SYSTEMS AND METHODS,” which is now issued as U.S. Pat. No. 11,154,314, which claims the benefit of U.S. Provisional Patent Application No. 62/622,691, filed on Jan. 26, 2018, and titled “SINGLE INSERTION DELIVERY SYSTEM FOR TREATING EMBOLISM AND ASSOCIATED SYSTEMS AND METHODS,” each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present technology relates generally to devices and methods for the intravascular treatment of emboli and/or thrombi within a blood vessel of a human patient. In particular, some embodiments of the present technology relate to systems for repeatedly deploying an interventional device at or proximate to a pulmonary embolism within a patient.

BACKGROUND

[0003] Thromboembolic events are characterized by an occlusion of a blood vessel.

Thromboembolic disorders, such as stroke, pulmonary embolism, heart attack, peripheral thrombosis, atherosclerosis, and the like, affect many people. These disorders are a major cause of morbidity and mortality.

[0004] When an artery is occluded by a clot, tissue ischemia develops. The ischemia will progress to tissue infarction if the occlusion persists. Infarction does not develop or is greatly limited if the flow of blood is reestablished rapidly. Failure to reestablish blood flow can lead to the loss of limb, angina pectoris, myocardial infarction, stroke, or even death.

[0005] In the venous circulation, occlusive material can also cause serious harm. Blood clots can develop in the large veins of the legs and pelvis, a common condition known as deep venous thrombosis (DVT). DVT arises most commonly when there is a propensity for stagnated blood (e.g., long distance air travel, immobility, etc.) and clotting (e.g., cancer, recent surgery, such as orthopedic surgery, etc.). DVT causes harm by: (1) obstructing drainage of venous blood from the legs leading to swelling, ulcers, pain, and infection, and (2) serving as a reservoir for blood clots to travel to other parts of the body including the heart, lungs, brain (stroke), abdominal organs, and/or extremities.

[0006] In the pulmonary circulation, the undesirable material can cause harm by obstructing pulmonary arteries—a condition known as pulmonary embolism. If the obstruction is upstream, in the main or large branch pulmonary arteries, it can severely compromise total blood flow within the lungs, and therefore the entire body, and result in low blood pressure and shock. If the obstruction is downstream, in large to medium pulmonary artery branches, it can prevent a significant portion of the lung from participating in the exchange of gases to the blood resulting in low blood oxygen and buildup of blood carbon dioxide.

[0007] There are many existing techniques to reestablish blood flow through an occluded vessel. One common surgical technique, an embolectomy, involves incising a blood vessel and introducing a balloon-tipped device (such as the Fogarty catheter) to the location of the occlusion. The balloon is then inflated at a point beyond the clot and used to translate the obstructing material back to the point of incision. The obstructing material is then removed by the surgeon. Although such surgical techniques have been useful, exposing a patient to surgery may be traumatic and best avoided when possible. Additionally, the use of a Fogarty catheter may be problematic due to the possible risk of damaging the interior lining of the vessel as the catheter is being withdrawn.

[0008] Percutaneous methods are also utilized for reestablishing blood flow. A common percutaneous technique is referred to as balloon angioplasty where a balloon-tipped catheter is introduced to a blood vessel (e.g., typically through an introducing catheter). The balloon-tipped catheter is then advanced to the point of the occlusion and inflated to dilate the stenosis. Balloon angioplasty is appropriate for treating vessel stenosis, but it is generally not effective for treating acute thromboembolisms as none of the occlusive material is removed and the vessel will re-stenosis after dilation. Another percutaneous technique involves placing a catheter near the clot and infusing streptokinase, urokinase, or other thrombolytic agents to dissolve the clot. Unfortunately, thrombolysis typically takes hours to days to be successful. Additionally, thrombolytic agents can cause hemorrhage and in many patients the agents cannot be used at all.

[0009] Various devices exist for performing a thrombectomy or removing other foreign material. However, such devices have been found to have structures which are either highly complex, cause trauma to the treatment vessel, or lack sufficient retaining structure and thus cannot be appropriately fixed against the vessel to perform adequately. Furthermore, many of the devices have highly complex structures that lead to manufacturing and quality control difficulties as well as delivery issues when passing through tortuous or small diameter catheters. Less complex devices may allow the user to pull through the clot, particularly with inexperienced users, and such devices may not completely capture and/or collect all of the clot material.

[0010] Moreover, with many devices, it is difficult or not possible to make repeated attempts at removing clot material (e.g., to make multiple passes with a device). In particular, if a first pass with a device does not completely capture and/or collect all of the clot material, the device and an accompanying catheter system must be removed from the patient, cleaned, and subsequently reinserted into the patient in order to make a second pass and remove additional material. This can be time consuming and traumatic for the patient.

[0011] Thus, there exists a need for an improved embolic extraction device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Many aspects of the present technology can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present disclosure.

[0013] FIGS. 1A and 1B are perspective views of a retraction and aspiration system shown in a first state and a second state, respectively, in accordance with embodiments of the present technology.

[0014] FIGS. 2A-2G are schematic illustrations of a distal portion of the retraction and aspiration system during a clot removal procedure in accordance with embodiments of the present technology.

[0015] FIGS. 3A-3C are a fully-exploded view, an isometric view, and a partially-exploded view of a clot reservoir of the retraction and aspiration system in accordance with embodiments of the present technology.

[0016] FIG. 4 is a side view of an attachment member and a fluid control unit of the retraction and aspiration system in accordance with embodiments of the present technology.

[0017] FIG. 5 is a side cross-sectional view of the attachment member shown in FIG. 4.

[0018] FIG. 6 is an isometric view of a first valve insert in accordance with embodiments of the present technology.

[0019] FIGS. 7A and 7B are a side view and a side cross-sectional view, respectively, of the first valve insert shown in FIG. 6 inserted into the attachment member shown in FIG. 4 in accordance with embodiments of the present technology.

[0020] FIG. 8 is a side view of a second valve insert in accordance with embodiments of the present technology.

[0021] FIG. 9 is a side view of the second valve insert shown in FIG. 8 inserted into the attachment member shown in FIG. 4 in accordance with embodiments of the present technology.

[0022] FIG. 10 is a flow diagram of a process or method for operating the retraction and aspiration system in accordance with embodiments of the present technology.

[0023] FIG. 11 is a side view of an attachment member and a fluid control unit of the retraction and aspiration system in accordance with embodiments of the present technology.

[0024] FIG. 12 is a side view of the fluid control unit shown in FIG. 11 in accordance with embodiments of the present technology.

[0025] FIGS. 13A and 13B are side cross-sectional views of the attachment member shown in FIG. 11 in a first configuration and a second configuration, respectively, in accordance with embodiments of the present technology.

[0026] FIG. 14 is a flow diagram of a process or method for operating the retraction and aspiration system in accordance with embodiments of the present technology.

DETAILED DESCRIPTION

[0027] The present technology is generally directed to systems and associated devices and methods for engaging and removing clot material from a blood vessel of a human patient. In some embodiments, an interventional device can be advanced through a guide catheter and deployed within clot material in a blood vessel. The interventional device can subsequently be withdrawn from the patient through the guide catheter to remove clot material captured by the interventional device. In some embodiments of the present technology, the interventional device can be repeatedly deployed in/withdrawn from the blood vessel to capture a desired amount of the clot material—without requiring that the guide catheter be fully withdrawn from the patient after each “pass” (e.g., each repeated deployment/withdrawal of the interventional device). That is, the guide catheter may be inserted only a single time during an intravascular procedure including multiple passes to remove clot material from the patient.

[0028] Although many of the embodiments are described below with respect to devices, systems, and methods for treating a pulmonary embolism, other applications and other embodiments in addition to those described herein are within the scope of the technology (e.g., intravascular procedures other than the treatment of emboli, intravascular procedures for treating cerebral embolism, etc.). Additionally, several other embodiments of the technology can have different configurations, states, components, or procedures than those described herein. Moreover, it will be appreciated that specific elements, substructures, advantages, uses, and/or other features of the embodiments described with reference to FIGS. **1A-14** can be suitably interchanged, substituted or otherwise configured with one another in accordance with additional embodiments of the present technology. Furthermore, suitable elements of the embodiments described with reference to FIGS. **1A-14** can be used as standalone and/or self-contained devices. A person of ordinary skill in the art, therefore, will accordingly understand that the technology can have other embodiments with additional elements, or the technology can have other embodiments without several of the features shown and described below with reference to FIGS. **1A-14**.

[0029] With regard to the terms “distal” and “proximal” within this description, unless otherwise specified, the terms can reference a relative position of the portions of a retraction and aspiration apparatus and/or an associated catheter system with reference to an operator and/or a location in the vasculature. Also, as used herein, the designations “rearward,” “forward,” “upward,” “downward,” etc. are not meant to limit the referenced component to use in a specific orientation. It will be appreciated that such designations refer to the orientation of the referenced component as illustrated in the Figures; the retraction and aspiration system of the present technology can be used in any orientation suitable to the user.

I. SELECTED EMBODIMENTS OF RETRACTION/ASPIRATION SYSTEMS AND METHODS OF USE

[0030] FIGS. **1A** and **1B** are perspective views of a proximal portion of a clot retrieval system **1** configured in accordance with embodiments of the present technology, shown in a first state and a second state, respectively. Referring to FIGS. **1A** and **1B** together, the clot retrieval system **1** includes a retraction and aspiration device **100** (“RA device **100**”), a catheter system **200**, and a tubing system **300**. In some embodiments, the RA device **100** is coupleable to the catheter system **200** and operable to simultaneously (i) retract a portion of the catheter system **200** and (ii) aspirate through the catheter system **200**. In certain embodiments, the RA device **100** and catheter system **200** can both be fluidly coupled to the tubing system **300** to enable material (e.g., blood and clot material) aspirated from the catheter system **200** to flow into the tubing system **300**. In particular, the RA device **100**, the catheter system **200**, and the tubing system **300** can be the same as or similar to one or more of the retraction and aspiration devices, catheter systems, and tubing systems disclosed in U.S. Pat. No. 9,526,864, filed Jun. 9, 2015, and titled “RETRACTION AND ASPIRATION DEVICE FOR TREATING EMBOLISM AND ASSOCIATED METHODS,” which is incorporated herein by reference in its entirety.

[0031] FIGS. **2A-2G** are schematic illustrations of a distal portion of the clot retrieval system **1** during a clot removal procedure in accordance with embodiments of the present technology. In particular, FIGS. **2A-2G** illustrate a distal portion of the catheter system **200** that is positioned proximate to an embolism or clot material within a blood vessel (e.g., a pulmonary blood vessel). Accordingly, operation of the catheter system **200** is described with references to FIGS. **2A-2G**.

[0032] FIG. **2A** is a side view of a distal portion of the catheter system **200** positioned adjacent an embolism or clot material PE within a blood vessel BV (e.g., a pulmonary blood vessel). As shown in FIG. **2A**, the catheter system **200** can include an outer guide catheter **206** defining a lumen **205**, a delivery sheath **204** slidably received within the lumen of the guide catheter **206**, and an elongated pull (and/or push) member **202** slidably received within a lumen of the delivery sheath **204**. The guide catheter **206** and the delivery sheath **204** individually comprise an elongated shaft having a lumen and, in some embodiments, the push member **202** can also define a lumen (e.g.,

configured to receive a guidewire therethrough). In a particular embodiment, the catheter system **200** does not include a guide catheter **206** and/or a delivery sheath **204**. As shown in FIG. 2A, a distal portion of the push member **202** can be integral with or coupled to an interventional device ID, such as a clot removal and/or clot treatment device, that is housed within the delivery sheath **204**. Accordingly, axial movement of the pull member **202** causes axial movement of the interventional device ID.

[0033] As further shown in FIG. 2A, the delivery sheath **204** and the interventional device ID can be positioned at least partially within the clot material PE. Access to the pulmonary vessels can be achieved through the patient's vasculature, for example, via the femoral vein. The catheter system **200** can include an introducer **210** (FIGS. 1A and 1B; e.g., a Y-connector with a hemostasis valve) that can be partially inserted into the femoral vein. A guidewire (not shown) can be guided into the femoral vein through the introducer **210** and navigated through the right atrium, the tricuspid valve, the right ventricle, the pulmonary valve and into the main pulmonary artery. Depending on the location of the embolism, the guidewire can be guided to one or more of the branches of the right pulmonary artery and/or the left pulmonary artery. It will be understood, however, that other access locations into the venous circulatory system of a patient are possible and consistent with the present technology. For example, the user can gain access through the jugular vein, the subclavian vein, the brachial vein or any other vein that connects or eventually leads to the superior vena cava. Use of other vessels that are closer to the right atrium of the patient's heart can also be advantageous as it reduces the length of the instruments needed to reach the pulmonary embolism.

[0034] As shown in FIGS. 2B and 2C, the delivery sheath **204** can be withdrawn proximally (e.g., as indicated by arrow A1 in FIG. 2B) to allow the interventional device ID to expand within the clot material PE, thereby grabbing the clot material PE that is nearby. Although FIG. 2B shows the interventional device ID positioned at the treatment site such that a distal terminus of the interventional device ID is distal of a distal terminus of the clot material PE, in some procedures the interventional device ID may be positioned such that the distal terminus of the interventional device ID is proximal of the distal terminus of the clot material PE. As shown in FIG. 2D, in some embodiments the guide catheter **206** can optionally be advanced distally (e.g., as indicated by arrow A2) until the guide catheter **206** is positioned proximate to a proximal portion of the clot material PE.

[0035] Referring again to FIGS. 1A and 1B, the RA device **100** includes (i) a housing **102**, (ii) an actuation mechanism that includes a lever **104** coupled to and extending from the housing **102**, and (iii) a pressure source (obscured in FIGS. 1A and 1B; e.g., a syringe) positioned within the housing **102**, coupled to the actuation mechanism, and configured to generate negative and/or positive pressure. The lever **104** is shown in a first position and second position in FIGS. 1A and 1B, respectively. The housing **102** can have a proximal portion **100a**, a distal portion **100b**, and an opening **114** at the distal portion **100b** configured to receive a portion of the catheter system **200** and to mechanically couple the catheter system **200** to the housing **102**. For example, a proximal portion of the guide catheter **206** can include an attachment/valve member **208** that is configured to be detachably coupled to the RA device **100** (e.g., via a snap-fit arrangement) to secure the catheter **200** to the RA device **100**. As described in greater detail below, the attachment/valve member **208** can fluidly couple an aspiration lumen of the catheter system **200** (e.g., the lumen **205** of the guide catheter **206**) to the tubing system **300** of the clot retrieval system **1**.

[0036] The housing **102** can further include a channel **116** that extends proximally from the opening **114** along approximately the length of the housing **102**, as shown in FIGS. 1A and 1B. The channel **116** can have a height at least as great as the outer diameter of the delivery sheath **204** of the catheter system **200** (and/or another component of the catheter system **200**) such that the delivery sheath **204** can fit sideways through the channel **116**. In some embodiments, the pull member **202** and the interventional device ID (FIGS. 2A-2G) can be pre-loaded into the delivery sheath **204**, and the delivery sheath **204** can be fed distally through the channel **116** (e.g., either via

the proximal end of the channel **116** or first pushed sideways through a portion of the channel **116**) and into the guide catheter **206**. In other embodiments, the interventional device ID and the delivery sheath **204** are fed into the guide catheter **206** and the interventional device ID is deployed prior to coupling the guide catheter **206** to the RA device **100**.

[0037] When the RA device **100** is coupled to the catheter system **200** (e.g., when the attachment/valve member **208** of the catheter system **200** is positioned within the opening **114** in the housing **102** of the RA device **100**), movement of the lever **104** functions to retract a portion of the catheter system **200** positioned in the channel **116** (e.g., the delivery sheath **204** and/or the push member **202**). For example, the RA device **100** can include a locking portion that grips the delivery sheath **204** (and, in some embodiments, indirectly the push member **202**) to pull the delivery sheath **204** proximally as the lever **104** is moved from the first to the second position.

[0038] The tubing system **300** of the clot retrieval system **1** fluidly couples the pressure source of the RA device **100** to the aspiration lumen of the catheter system **200**. When the RA device **100** is coupled to the catheter system **200**, movement of the lever **104** functions to simultaneously generate negative pressure in the pressure source and to retract a portion of the catheter system **200**, as described above. The tubing system **300** has a first portion **314** coupled to the pressure source, a second portion **316** coupled to the guide catheter **206**, and a drainage portion **318** coupled to a reservoir **320** (e.g., a vinyl bag). The first portion **314**, second portion **316**, and/or drainage portion **318** can include one or more tubing sections **302** (labeled individually as tubing sections **302a-302f**) and/or fluid control unit, such as one or more control valves. In certain embodiments, one or more of the tubing sections **302** can have a relatively large diameter (e.g., greater than about 0.1 inch, greater than about 0.210 inch, etc.) to help inhibit clogging of clot material within the tubing system **300**.

[0039] More specifically, the first portion **314** can include the tubing section **302a**. The drainage portion **318** can include (i) the tubing section **302b**, (ii) a first fluid control unit (e.g., a valve) **304**, and (iii) the tubing section **302c**. The second portion **316** can include (i) the tubing section **302d**, (ii) a clot reservoir **306**, (iii) the tubing section **302e**, (iv) a second fluid control unit **310**, and (v) the tubing section **302f**. In some embodiments, the first fluid control unit **304** can be a one-way valve (e.g., a check valve) that only allows fluid flow from the first portion **314** and/or second portion **316** to the drainage portion **318** (and not vice-versa). In certain embodiments, as described in detail with reference to FIGS. 3A-3C, the clot reservoir **306** can also include a one-way valve that only allows fluid flow from the second portion **316** to the drainage portion **318** (and not vice-versa). In some embodiments, the second fluid control unit **310** can be a stopcock or a clamp that is externally operated to regulate the flow of liquid through the second portion **316** of the tubing system **300**. A Y-connector **308** can fluidly couple the first, second, and drainage portions **314**, **316**, **318**. In other embodiments, the first, second, and/or drainage portions **314**, **316**, **318** can have more or fewer tubing sections, connectors, and/or fluid control unit and/or other suitable configurations.

[0040] As shown in FIGS. 1A and 2E, moving the lever **104** from the first position to the second position (indicated by arrow A1 in FIG. 1A) simultaneously (1) generates a negative pressure in the lumen **205** of the guide catheter **206** (indicated by arrow F in FIG. 1A and F1 in FIG. 2E), and (2) retracts the delivery sheath **204** and/or push member **202** proximally, thereby retracting the interventional device ID from the treatment site. During this time, in some embodiments, the guide catheter **206** remains fixed (e.g., by the housing **102**) relative to the delivery sheath **204** and pull member **202**. In such embodiments, as the lever **104** moves from the first position to the second position, the interventional device ID, delivery sheath **204**, pull member **202**, and at least a portion of the clot material PE are drawn proximally into the guide catheter **206**.

[0041] As shown in FIG. 1B, moving the lever **104** from the second position to the first position (indicated by arrow A2 in FIG. 1B) creates a positive pressure (e.g., indicated by arrows F in FIG. 1B) in the first portion **314** and drainage portion **318** of the tubing system **300**. The clot reservoir **306** prevents the positive pressure from affecting the aspiration lumen of the catheter system **200**,

thereby preventing backflow of fluid into the blood vessel BV at the treatment site. With respect to the catheter system **200**, when the lever **104** is actuated from the second position to the first position, the RA device **100** does not engage the delivery sheath **204** or the push member **202** to move these components. Thus, the next time the lever **104** is actuated relative to the housing **102**, the RA device **100** engages a new portion of the delivery sheath **204** and push member **202** such that the delivery sheath **204** and push member **202** are incrementally retracted proximally each time the lever **104** is “pumped” (e.g., moved from the first position toward the second position and then back toward the first position).

[0042] Depending on the age and size of the clot material PE, local anatomical and/or physiological conditions, and position of the interventional device ID relative to the clot material PE, the lever **104** can be pumped several times to fully extract the clot material PE and/or interventional device ID from the treatment site. For example, FIGS. 2D and 2E show the proximal movement of the delivery sheath **204** and pull member **202** after a first pump of the lever **104**. FIGS. 2E and 2F show the proximal movement of the delivery sheath **204**, pull member **202**, and interventional device ID after a second pump of the lever **104** (e.g., including a second instance of pressure generation indicated by arrows F1 and F2 in FIGS. 2E and 2F, respectively). In some embodiments, the interventional device ID and the clot material PE can be fully withdrawn into the guide catheter **206** after a single pump of the lever **104**. In other embodiments, such as those procedures where the interventional device ID is initially positioned such that a distal terminus of the interventional device ID is proximal of a distal terminus of the clot material PE (the clot material PE often originates in a vein of the patient's leg, and thus is cast into an elongated, worm-like shape), it can take several pumps of the lever **104** to fully withdraw the clot material PE into the guide catheter **206**. Thus, in some embodiments, even when the interventional device ID is positioned within the guide catheter **206** such that a distal terminus of the interventional device ID is proximal of the distal terminus of the guide catheter **206**, the lever **104** can be pumped several more times to continue to withdraw the clot material PE into the guide catheter **206** and the tubing system **300** (e.g., into the second portion **316**, the clot reservoir **306**, and/or the drainage portion **318**).

[0043] Once the clot material PE is positioned within the guide catheter **206** such that a distal terminus of the clot material PE is proximal from a distal terminus of the guide catheter **206**, the catheter system **200** can be withdrawn proximally (e.g., as indicated by arrow A3 in FIG. 2G) from the treatment site, and removed from the patient. However, sometimes, as shown in FIG. 2G, retracting the interventional device ID and delivery sheath **204** into the guide catheter **206** may not remove all of the clot material PE (or a desired amount of the clot material PE) from the blood vessel BV. That is, a single “pass” (e.g., a deployment of the interventional device ID and subsequent retraction of the interventional device ID into the guide catheter **206**) may not adequately remove the clot material PE from the blood vessel BV. In such instances, the operator of the clot retrieval system **1** may wish to make another pass with the interventional device ID to remove all or a portion of the remaining clot material PE in the blood vessel BV.

[0044] To redeploy the interventional device ID, many conventional systems require that the entire catheter system, including the guide catheter **206**, be fully removed from the patient (e.g., including a guide catheter). That is, if the once-deployed interventional device is reintroduced without fully removing and cleaning the catheter system, there is a significant risk that clot material and/or other contaminants from the catheter system will be reintroduced into the blood vessel of the patient during a second pass. As described in further detail below, the present technology advantageously allows for an interventional device to be redeployed without fully removing a guide catheter, and with a significantly reduced risk of reintroducing clot material and/or other contaminants into the blood vessel of the patient.

II. SELECTED EMBODIMENTS OF CLOT RESERVOIRS AND ASSOCIATED METHODS OF USE

[0045] FIGS. 3A-3C are a fully-exploded view, an isometric view, and a partially-exploded view of

the clot reservoir **306** of the clot retrieval system **1**. With reference to FIG. 3A, the clot reservoir **306** includes a housing **322** defining a chamber **323**, a filter **321** configured to be positioned within the housing **322**, and a cap assembly **330** configured to be coupled to the housing **322**. The housing **322** can include a port **324** configured to be removably, fluidly coupled to the catheter system **200**. In some embodiments, a tubing section **326** is coupled to the port **324** during priming of the clot retrieval system **1** and, as described in further detail below with reference to FIGS. **10** and **14**, during flushing of the tubing system **300**. In certain embodiments the tubing section **326** is semi-permanently coupled to the housing **322** and can be used to fluidly couple the housing **322** to the tubing system **300** and/or the catheter system **200**.

[0046] In the embodiment illustrated in FIG. 3A, the cap assembly **330** includes a fluid connector (e.g., a barbed outlet) **332** for connecting to the tubing system **300** (e.g., to the tubing section **302d** shown in FIGS. **1A** and **1B**), a nut **333** for releasably securing the cap assembly **330** to the housing **322** (e.g., via a threaded coupling), a first cap portion **334**, and a second cap portion **338**. In some embodiments, the fluid connector **332** is secured to the first cap portion **334** via an adhesive **331a**, and the first cap portion **334** is secured to the nut **333** via a second adhesive **331b**. In some embodiments, the first and second adhesives **331a**, **331b** are the same. The cap assembly **330** can further include a check valve assembly **335** positioned between the first and second cap portions **334**, **338** and comprising a piston **336** and a spring **339** (e.g., a passivated compression spring). The check valve assembly **335** provides for one-way fluid flow through the clot reservoir **306**, for example, from the catheter system **200** to the reservoir **320** (FIGS. **1A** and **1B**). The clot reservoir **306** can further include one or more O-rings **337** for sealing the various components.

[0047] In operation, blood and clot material flow into the clot reservoir **306** via the port **324** as the lever **104** of the RA device **100** is moved from the first position toward the second position. Clot material is captured within the housing **322** and inhibited from exiting through the cap assembly **330** by the filter **321** while blood is allowed to flow from the port **324** to the fluid connector **332**. In particular, the filter **321** inhibits clot material from passing into the check valve assembly **335**, which could inhibit function of the check valve assembly **335** and/or macerate the clot material and make it indistinguishable from or difficult to distinguish from other fluids (e.g., blood) aspirated and/or removed from the patient. The check valve assembly **335** subsequently inhibits backflow of fluid through the housing **322** via the fluid connector **332** as the lever **104** of the RA device **100** is moved from the second position toward the first position. With reference to FIGS. **3B** and **3C**, the cap assembly **330** can be decoupled from the housing **322** to, for example, permit an operator to remove clot material collected in the housing **322**. Moreover, as shown, the housing **322** may be made of a transparent material that permits the operator to visualize material within the housing **322**. As described in further detail below, in some embodiments, the operator can at least partially determine whether subsequent passes using the interventional device **100** are necessary by visualizing the amount of clot material collected in the housing **322**.

[0048] Additional details of the clot reservoir **306**, and associated devices and methods, are described in Appendix A to this application.

III. SELECTED EMBODIMENTS OF ATTACHMENT/VALVE MEMBERS, AND ASSOCIATED DEVICES AND METHODS OF USE

[0049] FIG. **4** is a side view of an attachment/valve member **408** (“attachment member **408**”) of the catheter system **200** and a second fluid control unit **410** of the tubing system **300** in accordance with embodiments of the present technology. The attachment member **408** and second fluid control unit **410** can have some features generally similar to the features of the attachment/valve member **208** and second fluid control unit **310**, respectively, described above with reference to FIGS. **1A** and **1B**. For example, the attachment member **408** can be integral with or coupled to a proximal portion of the guide catheter **206** and configured to be detachably coupled to the RA device **100** (FIGS. **1A** and **1B**; e.g., via a snap-fit arrangement) to at least partially secure the catheter system **200** to the RA device **100**. When secured to the RA device **100**, the attachment member **408** can

fluidly couple the lumen **205** (e.g., an aspiration lumen) of the guide catheter **206** to the tubing system **300** of the clot retrieval system **1** via the tubing section **302f**. Likewise, the second fluid control unit **410** can be a clamp (“clamp **410**”) that is externally operable to regulate the flow of liquid through the tubing section **302f**. For example, the clamp **410** may be actuated (e.g., compressed or squeezed by the hand of an operator) to partially or fully restrict fluid flow through the tubing section **302f**. In some embodiments, the clamp **410** includes features for locking or maintaining the clamp **410** in a position such that it restricts fluid flow through the tubing section **302f**. FIG. 4 illustrates the clamp **410** in a position that permits fluid flow through the tubing section **302f**.

[0050] As further shown in FIG. 4, the tubing system **300** can include a connector **417** that, for example, fluidly connects the tubing section **302f** to other portions of the tubing system **300** (e.g., those shown in FIGS. 1A and 1B). In some embodiments, the connector **417** is a quick-release connector that enables rapid coupling/decoupling of the tubing section **302f** to the clot reservoir **306** and/or other components of the tubing system **300**. In some embodiments, the tubing system **300** can include a flush port adapter that can be removably coupled to the connector **417**. The flush port adapter can be configured to, for example, fluidly connect a flushing device (e.g., a syringe) to the catheter system **200** so that the guide catheter **206** can further be flushed with a fluid (e.g., heparinized saline).

[0051] FIG. 5 is a side cross-sectional view of the attachment member **408** shown in FIG. 4. As shown in the embodiment of FIG. 5, the attachment member **408** includes a body or housing **440** having a proximal opening **443a**, a distal opening **443b**, and a first lumen **441** extending between the proximal and distal openings **443a** and **443b**. The first lumen **441** further comprises a first portion **441a** having a first diameter (e.g., a constant first diameter) and a second portion **441b** having a second diameter greater than the first diameter. In some embodiments, the first diameter can be generally the same as the outer diameter of the guide catheter **206**. The housing **443** further includes a branch portion **444** configured to be coupled to the tubing section **302f** and having a second lumen **442** branching from the first lumen **441**. In some embodiments, the second lumen **442** can have a relatively large diameter (e.g., between about 0.098 inch and 0.210 inch, about 0.210 inch, greater than 0.210 inch, etc.) to help inhibit clogging and/or collecting of clot material within the attachment member **408** during, for example, aspiration of the guide catheter **206**.

[0052] As illustrated in the embodiment of FIG. 5, the attachment member **408** can further include a valve **445** within the second portion **441b** of the first lumen **441**. The valve **445** can be, for example, a hemostasis valve configured to maintain hemostasis by preventing fluid flow in the proximal direction through the proximal opening **443a** of the attachment member **408** as the delivery sheath **204**, the pull member **202**, the guidewire, the interventional device ID, and/or other components of the catheter system **200** are inserted through the valve **445** for advancement to the treatment site in the blood vessel BV.

[0053] FIG. 6 is an isometric view of a first valve insert **650**, such as a hub valve insert, in accordance with embodiments of the present technology. The first valve insert **650** is configured to be inserted at least partially into the first lumen **441** of the attachment member **408** (e.g., into the valve **445**) through the proximal opening **443a** of the attachment member **408**. Accordingly, FIGS. 7A and 7B are a side view and a side cross-sectional view, respectively, of the first valve insert **650** inserted into the attachment member **408** in accordance with embodiments of the present technology.

[0054] Referring to FIG. 6, the first valve insert **650** includes a proximal portion **651a**, a distal portion **651b** extending from the proximal portion **651a**, and a lumen **654**. As illustrated in the embodiment of FIG. 6, the proximal portion **651a** can optionally include one or more first engagement features (e.g., flanges, tabs, etc.) **652** configured to engage with the attachment member **408** of the catheter system **200** to securely position (e.g., lock, mate, flush, etc.) the first valve insert **650** within the attachment member **408**. For example, in some embodiments, the

first engagement features **652** can be configured to “snap” into (e.g., mate with) corresponding grooves on the attachment member **408**. The proximal portion **651a** can further include one or more second engagement features **653** (e.g., flanges, tabs, etc.) configured to be gripped by an operator to enable the operator to, for example, easily manipulate and/or position the first valve insert **650** within the attachment member **408**.

[0055] Referring to FIGS. **6-7B** together, the distal portion **651b** of the first valve insert **650** is configured to be positioned at least partially within the second portion **441b** of the first lumen **441** of the attachment member **408**. For example, the first valve insert **650** can be advanced distally over the delivery sheath **204** and/or the guidewire of the catheter system **200** to the attachment member **408**. Once inserted into the attachment member **408**, the first valve insert **650** opens (e.g., exercises) the valve **445**. In some embodiments, the lumen **654** of the distal portion **651b** of the first valve insert **650** has a generally constant diameter that is generally the same as the first diameter of the first portion **441a** of the first lumen **441** of the attachment member **408**. In the embodiment illustrated in FIG. **7B**, when the first valve insert **650** is within the attachment member **408**, the distal portion **651b** extends substantially or entirely through the second portion **441b** of the first lumen **441** such that the lumen **654** of the first valve insert **650** and the second portion **441a** of the first lumen **441** of the attachment member **408** together define a generally continuous lumen extending through the attachment member **408** (e.g., between the distal opening **443b** of the attachment member **408** and a proximal terminus of the lumen **654**).

[0056] In one aspect of the present technology, the continuous lumen formed by inserting the first valve insert **650** into the attachment member **408** can have a generally constant diameter along the length of the lumen configured to accommodate the outer diameter of the guide catheter **206**. Accordingly, as described in further detail below, when a clot retaining portion of an interventional device and associated clot material are retracted proximally through the guide catheter **206**, the interventional device ID does not greatly change shape (e.g., expand or compress) while passing through the attachment member **408** and the likelihood of clot material being retained within the attachment member **408** is greatly reduced.

[0057] FIG. **8** is a side view of a second valve insert **860** (e.g., an aspiration insert) in accordance with embodiments of the present technology. Similar to the first valve insert **650**, the second valve insert **860** is configured to be inserted at least partially into the first lumen **441** of the attachment member **408** (e.g., through the valve **445**) through the proximal opening **443a** of the attachment member **408**. Accordingly, FIG. **9** is a side view of the second valve insert **860** inserted into the attachment member **408** in accordance with embodiments of the present technology.

[0058] Referring to FIG. **8**, the second valve insert **860** includes a proximal portion **861a**, a distal portion **861b** extending from the proximal portion **861a**, and a lumen **864** extending through the second valve insert **860**. As illustrated in the embodiment of FIG. **8**, the distal portion **861b** can optionally include one or more first engagement features (e.g., flanges, tabs, etc.) **862** configured to engage with the attachment member **408** to securely position (e.g., lock) the second valve insert **860** within the attachment member **408**. For example, in some embodiments, the first engagement features **862** can be configured to “snap” into (e.g., mate with) corresponding grooves on the attachment member **408**. The proximal portion **861a** can further include one or more second engagement features **865a** (e.g., flanges, tabs, etc.) configured to be gripped by an operator to enable the operator to, for example, easily manipulate and/or position the second valve insert **860** within the attachment member **408**.

[0059] The proximal portion **861a** can include one or more adjustment features **865** (labeled individually as adjustment features **865a** and **865b**) for adjusting a diameter of the lumen **864**. For example, in some embodiments the second valve insert **860** is a Tuohy Borst Adapter that can be adjusted, via the one or more adjustment features **865**, to seal the proximal opening **443a** of the attachment member **408** by sealing the lumen **864** against a component of the catheter system **200** inserted therethrough. More particularly, referring to FIGS. **8** and **9** together, the distal portion **861b**

of the second valve insert **860** is configured to be positioned at least partially within the attachment member **408** (e.g., within the second portion **441b** of the first lumen **441**). For example, the second valve insert **860** can be advanced distally over the guidewire of the catheter system **200** to the attachment member **408**. Once inserted into the attachment member **408**, the second valve insert **860** opens (e.g., exercises) the valve **445**. By tightening at least one of the adjustment features **865**, at least a portion of the lumen **864** can be narrowed until a seal is formed between the second valve insert **860** and the guidewire or other component of the catheter system **200** positioned therein.

[0060] As described in further detail below, in some embodiments, the second valve insert **860** may more completely seal against components of the catheter system **200** than the valve **445** of the attachment member **408**. Accordingly, use of the second valve insert **860** can improve the efficiency of aspiration of the guide catheter **206** using the RA device **100** (FIGS. **1A** and **1B**).

[0061] FIG. **10** is a flow diagram of a process or method **1000** for operating the clot retrieval system **1** including the attachment member **408** and the first and second valve inserts **650**, **860** to remove clot material from within a blood vessel (e.g., a pulmonary blood vessel) of a human patient in accordance with embodiments of the present technology. Although some features of the method **1000** are described in the context of the embodiments shown in FIGS. **1A-9** for sake of illustration, one skilled in the art will readily understand that the method **1000** can be carried out using other suitable systems and/or devices.

[0062] The method **1000** includes engaging the interventional device ID of the catheter system **200** with the clot material PE in the blood vessel BV as, for example, described above with reference to FIGS. **2A-2D** (block **1001**). In particular, the attachment member **408** (FIGS. **4** and **5**) of the catheter system **200** can be attached to the RA device **100**, and the interventional device ID can be deployed within at least a portion of the clot material PE by proximally retracting the delivery sheath **204**.

[0063] The method **1000** continues by proximally retracting the interventional device ID and associated clot material PE into the guide catheter **206** of the catheter system **200** until a distal terminus of the clot material PE is proximal from a distal terminus of the guide catheter **206** as, for example, described above with reference to FIGS. **2E-2G** (block **1002**). In particular, the RA device **100** can be pumped or cycled one or more times (e.g., one time, three times, five times, etc.) to retract the interventional device ID and/or delivery sheath **204** into the guide catheter **206** while simultaneously aspirating the lumen **205** of the guide catheter **206** to remove clot material PE and blood, which are drawn through the attachment member **408** and into the tubing system **300** (e.g., through the clot reservoir **306** to the reservoir **320**).

[0064] In some embodiments, the interventional device ID can be retracted proximally into the guide catheter **206** without use of the RA device **100**. For example, the operator can manually retract the interventional device ID and associated clot material PE into the guide catheter **206**.

[0065] After initial deployment of the interventional device ID in blocks **1001** and **1002**, the operator can determine whether it is necessary or desirable to redeploy the interventional device ID within the blood vessel BV of the patient in order to remove additional clot material PE that was not removed during a previous pass with the interventional device ID (block **1003**). In some embodiments, the operator can visualize the amount of clot material PE collected in the clot reservoir **306** to at least partially determine whether another pass is needed. In other embodiments, the operator can rely on imaging (e.g., fluoroscopic imaging) of the blood vessel BV or other techniques known in the art to determine whether an additional pass is necessary. If another pass is not needed (e.g., the clot material PE was adequately removed), the operator can elect to withdraw the catheter system **200** from the patient (block **1004**). If clot material PE remains in the vessel, the operator can prepare to redeploy the interventional device ID.

[0066] To redeploy the interventional device ID, the method **1000** includes positioning the first valve insert **650** (FIG. **6**) and withdrawing the interventional device ID from the patient (block **1005**). In particular, before positioning the first valve insert **650**, the catheter system **200** can be

decoupled from the RA device by (i) decoupling the attachment member **408** from the distal portion **100b** of the RA device **100** and (ii) removing the delivery sheath **204** and/or guidewire from the channel **116**. Additionally, in some embodiments, the clamp **410** can be actuated (e.g., closed) to prevent fluid flow from the guide catheter **206** into the tubing system **300** during withdrawal of the interventional device ID.

[0067] The first valve insert **650** can be advanced over the delivery sheath **204** and/or guidewire of the catheter system **200** until it is positioned (e.g., seated) at least partially within the attachment member **408** (FIGS. 7A and 7B). Once the first valve insert **650** is positioned within the attachment member **408**, the interventional device ID and delivery sheath can be fully withdrawn from the patient. For example, in some embodiments, the operator may grip a portion of the delivery sheath **204** that is exposed proximally from the RA device **100** and pull the delivery sheath **204** and interventional device ID proximally through the guide catheter **206**. In some embodiments, the first valve insert **650** is only positioned within the attachment member **408** when the interventional device ID is proximate the attachment member **408**. That is, the operator may partially withdraw the delivery sheath **204** and interventional device ID within the guide catheter **206** before advancing the first valve insert **650** into position. In some embodiments, the delivery sheath **204** includes a marking (e.g., a specific color pattern) configured to indicate to the operator that the distal end of the delivery sheath **204** and/or the interventional device ID is near (e.g., proximally approaching) the attachment member **408**. In certain embodiments, the guidewire of the catheter system **200** is pinned during withdrawal of the interventional device ID and the delivery sheath **204** such that the guidewire does not move relative to the interventional device ID and the delivery sheath **204**. Therefore, in such embodiments, the guidewire does not need to be re-advanced to the treatment site prior to an additional pass.

[0068] In operation, the first valve insert **650** helps create a lumen of constant diameter through the attachment member **408** such that a diameter of the interventional device ID does not substantially change (e.g., expand and/or contract) as the interventional device ID is withdrawn proximally through the attachment member **408**. In particular, the first valve insert **650** can effectively shield the interventional device ID from the valve **445** of the attachment member **408**. Without the first valve insert **650**, as the interventional device ID is withdrawn proximally it can expand within the attachment member **408** (e.g., as the interventional device ID passes through the second portion **441b** of the first lumen **441**) before being squeezed (e.g., radially collapsed) as it passes through the valve **445**. Without the first valve insert **650**, the valve **445** may strip (e.g., break off, shear, etc.) clot material PE that held by the interventional device ID. This can cause clot material PE to remain in the attachment member **408** after the interventional device ID is fully withdrawn from the patient, which presents a significant risk that remaining clot material PE will be reintroduced into the blood vessel BV of the patient if a second pass is made with the interventional device ID without fully removing the guide catheter **206** from the patient to enable cleaning of the guide catheter **206** and the attachment member **408**. The first valve insert **650** of the present technology inhibits clot material PE engaged with the interventional device ID from being stripped by the valve **445** within the attachment member **408** and, therefore, enables a second pass with the interventional device ID to be made without removing the guide catheter **206** from the patient.

[0069] The method **1000** includes removing the first valve insert **650** and positioning the second valve insert **860** (FIG. 9) (block **1006**). For example, the first valve insert **650** can be withdrawn proximally over the guidewire of the catheter system **200** and the second valve insert **860** can then be advanced distally over the guidewire until it is positioned within the attachment member **408**. In some embodiments, the first valve insert **650** is removed and the second valve insert **860** is inserted into the attachment member **408** immediately after the delivery sheath **204** and interventional device ID are withdrawn from the patient. Prompt removal of the first valve insert **650** can restore function of the valve **445** and prevent excess blood from leaking through the lumen **654** of the first valve insert **650** once the delivery sheath **204** and interventional device ID are removed.

[0070] Positioning the second valve insert **860** further includes adjusting (tightening) the second valve insert **860** over the guidewire to seal the proximal opening **443a** of the attachment member **408**. In some embodiments, the method **1000** need not include positioning of the second valve insert **860**. Instead, the valve **445** of the attachment member **408** may provide a suitable seal for subsequent aspiration steps. However, in some embodiments, use of the second valve insert **860** can provide a better seal between the guidewire of the catheter system **200** and the attachment member **408**, and thus improve the efficiency of aspiration using the RA device **100**.

[0071] The method **1000** includes aspirating the guide catheter **206** by, for example, pumping or cycling the lever **104** of the RA device **100** one or more times (block **1007**). In embodiments where the clamp **410** was previously closed (e.g., at block **1005**), prior to pumping the lever **104**, the operator can actuate (e.g., open) the clamp **410** to permit fluid flow from the guide catheter **206** into the tubing system **300**. Aspirating the guide catheter **206** removes any residual clot material PE remaining in the guide catheter **206**. Accordingly, the residual clot material PE is not reintroduced into the blood vessel BV of the patient when the interventional device ID and delivery sheath **204** are subsequently advanced through the guide catheter **206** during another pass. In certain embodiments, the guide catheter **206** can further be flushed with a fluid (e.g., heparinized saline). For example, the connector **417** can be decoupled from the tubing system **300** and the fluid can be introduced from a flushing device (e.g., a syringe) through a flush port adapter coupled (e.g., semi-permanently coupled) to the connector **417**.

[0072] In some embodiments, the guide catheter **206** can be aspirated without use of the RA device **100**. For example, a syringe or other pressure source can be fluidly coupled directly to the connector **417** and used to aspirate the guide catheter **206**. In such embodiments, opening of the clamp **410** fluidly connects the syringe to the lumen **205** of the guide catheter **206** and closing of the clamp **410** fluidly disconnects the syringe from the lumen **205** of the guide catheter **206**. In some embodiments, the syringe or other pressure source can be pre-charged with a vacuum—such as by drawing a plunger of the syringe with the clamp **410** closed. The clamp **410** can then be opened to instantaneously or nearly instantaneously (e.g., immediately) apply the stored vacuum pressure to the tubing system **300** and to the lumen **205** of the guide catheter **206**, thereby generating suction throughout the guide catheter **206**. In particular, suction can be generated at a distal portion of the guide catheter **206**. In one aspect of the present technology, pre-charging or storing the vacuum before applying the vacuum to the lumen **205** of the guide catheter **206** is expected to generate greater suction forces with a faster ramp time (and correspondingly greater fluid flow velocities) at and/or near a distal portion of the guide catheter **206** as compared to, for example, simply activating the pressure source of the RA device **100** by cycling the lever **104** of the RA device **100**. These suction forces generated by application of the stored vacuum can be used to not only aspirate the guide catheter **206**, but also to aspirate or otherwise remove some or all of the clot material PE remaining in the blood vessel BV after retraction of the interventional device ID.

[0073] The method **1000** includes flushing the tubing system **300** (block **1008**). In some embodiments, flushing the tubing system **300** includes (i) actuating (e.g., closing) the clamp **410** to inhibit fluid flow from the guide catheter **206** into the tubing system **300**, (ii) disconnecting the tubing section **326** of the clot reservoir **306** from the tubing section **302e** and the connector **417** of the tubing system **300**, (iii) placing the tubing section **326** into a container of fluid (e.g., saline), and (iv) pumping the lever **104** of the RA device **100** to draw the fluid through the tubing system **300**. In some embodiments, the housing **322** of the clot reservoir **306** can be temporarily disconnected (e.g., unscrewed) from the cap assembly **330** so that the clot material PE in the clot reservoir **306** can be removed. In certain embodiments, the tubing system **300** need not be flushed prior to a second pass with the interventional device ID or another interventional device. In some embodiments, flushing the tubing system **300** can include attaching a syringe to the fluid connector **332** of the clot reservoir **306** and/or to the tubing section **302a** of the tubing system **300** and using the syringe to generate a negative pressure to draw the fluid through the clot reservoir **306**.

[0074] After the tubing system **300** has been flushed, the method **1000** can return to block **1001**. In particular, the same interventional device ID and delivery sheath **204** can be cleaned and subsequently advanced through the guide catheter **206** and to the remaining clot material PE in the blood vessel BV. In some embodiments, a new interventional device and delivery sheath can be used for each pass to reduce the likelihood of contamination (e.g., reintroduction of clot material PE into the patient). Once the desired amount of clot material PE has been removed from the patient, the catheter system **200** may be fully withdrawn from the patient (block **1004**).

[0075] In one aspect of the present technology, the method **1000** provides for multiple passes of an interventional device without requiring that the entire guide catheter be removed after each pass. Accordingly, the present technology allows for only a single insertion of a guide catheter during a procedure including multiple passes to remove clot material-increasing the speed of the procedure and reducing trauma to the patient since the guide catheter does not need to be reintroduced (e.g., advanced through the vasculature and past the heart) before each pass.

[0076] Moreover, in certain embodiments, the present technology can enable the guide catheter **206** to be relocated to an alternate treatment site within the patient without removing the guide catheter **206** from the patient and, therefore, without reintroducing the guide catheter **206** through the heart. For example, the guide catheter **206** can be relocated to another treatment site within the lungs including a treatment site in the opposite lung. More specifically, (i) a dilator can be reintroduced into the guide catheter **206**, (ii) the guide catheter **206** can be withdrawn into the main pulmonary artery, (iii) the guidewire can be redirected to the new treatment site, (iv) the guide catheter **206** can be advanced over the guidewire to the new treatment site, and (v) the dilator can be removed.

[0077] Additional details of the systems, devices, and methods described above with reference to FIGS. **4-9** are provided in Appendix B to this application.

IV. OTHER SELECTED EMBODIMENTS OF ATTACHMENT/VALVE MEMBERS, AND ASSOCIATED DEVICES AND METHODS OF USE

[0078] FIG. **11** is a side view of an attachment/valve member **1108** (“attachment member **1108**”) of the catheter system **200** and a second fluid control unit **1110** of the tubing system **300** in accordance with embodiments of the present technology. The attachment member **1108** and the second fluid control unit **1110** can have some features generally similar to the features of the attachment members **208**, **408** and the second fluid control unit **310**, **410**, respectively, described above with reference to FIGS. **1A-10**. For example, the attachment member **1108** can be integral with or coupled to a proximal portion of the guide catheter **206** and configured to be detachably coupled to the RA device **100** (FIGS. **1A** and **1B**; e.g., via a snap-fit arrangement) to at least partially secure the catheter system **200** to the RA device **100**. When secured to the RA Device **100**, the attachment member **1108** can fluidly connect the lumen **205** (e.g., an aspiration lumen) of the guide catheter **206** to the tubing system **300** of the clot retrieval system **1** via the tubing section **302f**. Likewise, the second fluid control unit **1110** can be a stopcock (“stopcock **1110**”) that is externally operable to regulate the flow of fluid through the tubing section **302f**.

[0079] As further shown in FIG. **11**, the tubing system **300** can include a connector **1117** for, for example, fluidly connecting the tubing section **302f** and the stopcock **1110** to other portions of the tubing system **300** (e.g., those shown in FIGS. **1A** and **1B**). In some embodiments, the connector **1117** is a quick-release connector that enables rapid coupling/decoupling of the tubing section **302f** and stopcock **1110** to other components of the tubing system **300**.

[0080] FIG. **12** is a side view of the stopcock **1110** in accordance with embodiments of the present technology. In the embodiment illustrated in FIG. **12**, the stopcock **1110** includes a grip member (e.g., a handle, a knob, etc.) **1211** that can be actuated (e.g., twisted by the hand of an operator) to partially or fully restrict fluid flow through the tubing section **302f**. In some embodiments, the stopcock **1110** also includes an injection port **1212** (e.g., a needleless injection port) configured to receive a syringe or other fluid delivery member therethrough. In some embodiments, the injection port **1212** permits fluid to be introduced into the attachment member **1108** and the guide catheter

206 without requiring that a connector **1117** be decoupled from the tubing system **300**. For example, in some embodiments, an operator can actuate the stopcock **1110** (e.g., twist the grip member **1211**) to close the stopcock **1110** and subsequently introduce a syringe at the injection port **1212** to flush the guide catheter **206** with a fluid (e.g., saline) introduced via the syringe.

[0081] FIGS. **13A** and **13B** are side cross-sectional views of the attachment member **1108** in a first configuration and a second configuration, respectively, in accordance with embodiments of the present technology. The attachment member **1108** can be, for example, a garrote valve (e.g., a hemostasis valve) as disclosed in provisional U.S. Patent Application No. 62/554,931, filed Sep. 6, 2017, and titled “HEMOSTASIS VALVES AND METHODS OF USE,” which is reproduced in Appendix D to this application, and which is incorporated herein by reference in its entirety.

[0082] In particular, referring to FIGS. **13A** and **13B** together, the attachment member **1108** can include a housing **1370** having a proximal opening **1373a** and a distal opening **1373b**, and defining a first lumen (e.g., an interior channel) **1371** extending between the proximal and distal openings **1373a**, **1373b**. The housing **1370** further includes a branch portion **1380** configured to be coupled to the tubing section **302f** and defining a second lumen **1381** branching from the first lumen **1371**. In some embodiments, the second lumen **1381** can have a relatively large diameter (e.g., between about 0.098 inch and 0.210 inch, about 0.210 inch, greater than 0.210 inch, etc.) to help inhibit clogging and/or collection of clot material within the attachment member **1108** during, for example, aspiration of the guide catheter **206**. In one aspect of the present technology, the first lumen **1371** can have a generally constant diameter along its length.

[0083] As shown, a tubular member (e.g., an elongate member) **1372** can extend at least partially through the first lumen **1371** to define a central lumen **1374** that is generally coaxial with the first lumen **1371**. In some embodiments, the tubular member **1372** can comprise a compliant tubular structure (e.g., a silicon tube) that can be, for example, a thin-walled compliant tubular structure. The thin-walled structure of the tubular member **1372** can facilitate the collapse, and specifically the uniform collapse of the tubular member **1372** and sealing of the tubular member **1372**. For example, the attachment member **1108** can further include an actuation mechanism **1375** coupled to the tubular member **1372** and configured to collapse and seal the tubular member **1372** via compression and/or constriction of one or more filaments **1376** coupled to the tubular member **1372**.

[0084] More specifically, in some embodiments, the actuation mechanism **1375** can be a manual actuator such as one or more buttons **1378**. Depression or release of the buttons can, in some embodiments, facilitate sealing of the tubular member **1372** around tools or instruments of a wide range of sizes and/or diameters that fit through the tubular member **1372**. For example, FIG. **13A** illustrates the attachment member **1108** with the actuation mechanism **1375** in the first configuration in which the tubular member **1372** is in a collapsed and/or sealed state having a minimum diameter (e.g., conformed to a diameter of a portion of a catheter system inserted therethrough). In the embodiment shown in FIG. **13A**, the actuation mechanism **1375** is biased (e.g., by one or more springs **1377**) to maintain the tubular member **1372** in the collapsed state. Specifically, in the first configuration, the actuation mechanism **1375** can tighten the filament **1376** to constrict or compress the tubular member **1372** to seal the central lumen **1374** of the tubular member **1372**. Thus, when the buttons **1378** are not depressed, the attachment member **1108** can provide a hemostatic seal.

[0085] As shown in the embodiment of FIG. **13B**, the attachment member **1108** is in an expanded (e.g., open) and/or unsealed state having a maximum diameter (e.g., a diameter generally the same as the first lumen **1371**). More particularly, one or more of the buttons **1378** may be depressed to, for example, loosen the filament **1376** and allow expansion of the tubular member **1372** to unseal the central lumen **1374** of the tubular member **1372**.

[0086] Accordingly, the actuation mechanism **1375** and tubular member **1372** of the attachment member **1108** provide for sealing of the attachment member **1108** around, for example, various

components of the catheter system **200** (e.g., the delivery sheath **204**, the pull member **202**, the guidewire, the interventional device ID, etc.) that are inserted through the attachment member **1108** for advancement to the treatment site in the blood vessel BV. Moreover, in the second configuration, the central lumen **1374** of the tubular member **1372** and the first lumen **1371** of the housing **1370** can together provide a continuous lumen of generally constant diameter. As described above, such a constant diameter can prevent clot material PE associated with the interventional device ID from getting stuck in (e.g., remaining in) the attachment member **1108** as the interventional device ID is retracted through the attachment member **1108**—thus minimizing the risk of reintroducing clot material to the patient upon a second pass using the interventional device ID (or another interventional device).

[0087] FIG. **14** is a flow diagram of a process or method **1400** for operating the clot retrieval system **1** including the attachment member **1108** to remove clot material from within a blood vessel (e.g., a pulmonary blood vessel) of a human patient in accordance with embodiments of the present technology. Although some features of the method **1400** are described in the context of the embodiments shown in FIGS. **1A**, **1B**, and **1I-13** for sake of illustration, one skilled in the art will readily understand that the method **1400** can be carried out using other suitable systems and/or devices.

[0088] The method **1400** includes engaging the interventional device ID of the catheter system **200** with the clot material PE in the blood vessel BV as, for example, described above with reference to FIGS. **2A-2D** (block **1401**). In particular, the attachment member **1108** of the catheter system **200** can be coupled to the RA device **100**, and the interventional device ID can be deployed within at least a portion of the clot material PE by proximally retracting the delivery sheath **204** relative to the interventional device ID.

[0089] The method **1400** continues by proximally retracting the interventional device ID and associated clot material PE into the guide catheter **206** of the catheter system **200** until a distal terminus of the clot material PE is proximal from a distal terminus of the guide catheter **206** as, for example, described above with reference to FIGS. **2E-2G** (block **1402**). In particular, the RA device **100** can be pumped or cycled one or more times (e.g., one time, three times, five times, etc.) to retract the interventional device ID and/or delivery sheath **204** into the guide catheter **206** while simultaneously aspirating the lumen **205** of the guide catheter **206** to remove clot material PE and blood, which are drawn through the attachment member **1108** of the catheter system **200** and into the tubing system **300** (e.g., through the clot reservoir **306** to the reservoir **320**).

[0090] In some embodiments, the interventional device ID can be retracted proximally into the guide catheter **206** without use of the RA device **100**. For example, the operator can manually retract the interventional device ID and associated clot material PE into the guide catheter **206**.

[0091] After the initial deployment of the interventional device ID in blocks **1401** and **1402**, the operator can determine whether it is necessary or desirable to redeploy the interventional device ID within the blood vessel BV of the patient in order to remove additional clot material PE that was not removed during a previous pass with the interventional device ID (block **1403**). In some embodiments, the operator can visualize the amount of clot material PE collected in the clot reservoir **306** to at least partially determine whether another pass is needed. In other embodiments, the operator can rely on imaging (e.g., fluoroscopic imaging) of the blood vessel BV or other techniques known in the art to determine whether an additional pass is necessary. If another pass is not needed (e.g., the clot material PE was adequately removed), the operator can elect to withdraw the catheter system **200** from the patient at block **1404**. If clot material PE remains in the vessel, the operator can prepare to redeploy the interventional device ID.

[0092] To redeploy the interventional device ID, the method **1400** includes withdrawing the interventional device ID from the patient (block **1405**). In particular, before withdrawing the interventional device ID, the catheter system **200** can be decoupled from the RA device by (i) decoupling the attachment member **1108** from the distal portion **100b** of the RA device **100** and (ii)

removing the delivery sheath **204** and/or guidewire from the channel **116** of the RA device **100**. Additionally, in some embodiments, the stopcock **1110** can be actuated (e.g., twisted closed) to prevent fluid flow from the guide catheter **206** into the tubing system **300** during withdrawal of the interventional device ID.

[0093] The delivery sheath **204** and/or another component of the catheter system **200** may then be manually (e.g., by the operator) or automatically pulled proximally to withdraw the interventional device ID from the patient. Before the interventional device ID and associated clot material PE are withdrawn through the attachment member **1108**, the actuation mechanism **1375** of the attachment member **1108** can be actuated (e.g., by depressing the buttons **1378**) to move the tubular member **1372** to the second configuration (e.g., to open the tubular member **1372**) such that the central lumen **1374** of the tubular member **1372** and the first lumen **1371** of the housing **1370** together provide a continuous lumen of generally constant diameter. Accordingly, the interventional device ID can be fully withdrawn (e.g., retracted proximally) through the attachment member **1108** without causing a significant amount of clot material PE associated with the interventional device ID to remain in the attachment member **1108**—thus minimizing the risk of reintroducing clot material to the patient upon an additional pass using the interventional device ID (or another interventional device). Moreover, in certain embodiments, the guidewire of the catheter system **200** is pinned during withdrawal of the interventional device ID and the delivery sheath **204** such that the guidewire does not move relative to the interventional device ID and the delivery sheath **204**. Therefore, in such embodiments, the guidewire does not need to be re-advanced to the treatment site prior to an additional pass.

[0094] Once the interventional device ID has been fully removed from the guide catheter **206** and the attachment member **1108**, the attachment member **1108** can be returned to the first (e.g., sealed) configuration by, for example, releasing the buttons **1378**. Next, the method includes aspirating the guide catheter **206** by, for example, pumping or cycling the lever **104** of the RA device **100** one or more times (block **1406**). In embodiments where the stopcock **1110** was previously closed (e.g., at block **1405**), prior to pumping the lever **104**, the stopcock **1110** can be opened to permit fluid flow from the guide catheter **206** into the tubing system **300**. Aspirating the guide catheter **206** removes any residual clot material PE remaining in the guide catheter **206**. Accordingly, the residual clot material PE is not reintroduced into the blood vessel BV of the patient when the interventional device ID and delivery sheath **204** (or another interventional device ID) are subsequently advanced through the guide catheter **206** during another pass. In certain embodiments, the guide catheter **206** can further be flushed with a fluid (e.g., heparinized saline). For example, the fluid can be introduced through the injection port **1212** while simultaneously pressing the buttons **1378** of the attachment member **1108**. In certain embodiments, the stopcock **1110** can be closed (e.g., at block **1406**), and a syringe can be connected to the injection port **1212** and used to generate a negative pressure prior to opening the stopcock **1110** to permit fluid flow from the guide catheter **206** into the syringe.

[0095] In some embodiments, the guide catheter **206** can be aspirated without use of the RA device **100**. For example, a syringe or other pressure source can be fluidly coupled directly to the connector **1117** and used to aspirate the guide catheter **206**. In such embodiments, opening of the stopcock **1110** fluidly connects the syringe to the lumen **205** of the guide catheter **206** and closing of the stopcock **1110** fluidly disconnects the syringe from the lumen **205** of the guide catheter **206**. In some embodiments, the syringe or other pressure source can be pre-charged with a vacuum—such as by drawing a plunger of the syringe with the stopcock **1110** closed. The stopcock **1110** can then be opened to instantaneously or nearly instantaneously (e.g., immediately) apply the stored vacuum pressure to the tubing system **300** and to the lumen **205** of the guide catheter **206**, thereby generating suction throughout the guide catheter **206**. In particular, suction can be generated at a distal portion of the guide catheter **206**. In one aspect of the present technology, pre-charging or storing the vacuum before applying the vacuum to the lumen **205** of the guide catheter **206** is

expected to generate greater suction forces with a faster ramp time (and correspondingly greater fluid flow velocities) at and/or near a distal portion of the guide catheter **206** as compared to, for example, simply activating the pressure source of the RA device **100** by cycling the lever **104** of the RA device **100**. These suction forces generated by application of the stored vacuum can be used to not only aspirate the guide catheter **206**, but also to aspirate or otherwise remove some or all of the clot material PE remaining in the blood vessel BV after retraction of the interventional device ID. [0096] The method **1400** includes flushing the tubing system **300** (block **1407**). In some embodiments, flushing the tubing system **300** includes (i) closing the stopcock **1110** to inhibit fluid flow from the guide catheter **206** into the tubing system **300**, (ii) disconnecting the tubing section **326** of the clot reservoir **306** from the tubing section **300e** of the tubing system **300**, (iii) placing the tubing section **326** into a container of fluid (e.g., saline), and (iv) pumping the lever **104** of the RA device **100** to draw the fluid through the tubing system **300**. In some embodiments, the housing **322** of the clot reservoir **306** can be temporarily decoupled (e.g., unscrewed) from the cap assembly **330** so that the clot material PE in the clot reservoir **306** can be removed. In certain embodiments, the tubing system **300** need not be flushed prior to an additional pass with the interventional device ID. In some embodiments, flushing the tubing system **300** can include attaching a syringe to the fluid connector **332** of the clot reservoir **306** and/or to the tubing section **302a** of the tubing system **300** and using the syringe to generate a negative pressure to draw the fluid through the clot reservoir **306**.

[0097] After the tubing system **300** has been flushed, the method **1400** can return to block **1401**. In particular, the same interventional device ID and delivery sheath **204** can be cleaned and subsequently advanced through the guide catheter **206** and to the remaining clot material PE in the blood vessel. In some embodiments, a new interventional device ID and delivery sheath **204** can be used for each pass to reduce the likelihood of contamination (e.g., reintroduction of clot material PE). Once the desired amount of clot material PE has been removed from the patient, the catheter system **200** may be fully withdrawn from the patient (block **1404**).

[0098] In one aspect of the present technology, the method **1400** provides for multiple passes of an interventional device without requiring that the entire guide catheter be removed after each pass. Accordingly, the present technology allows for only a single insertion of a guide catheter during a procedure including multiple passes to remove clot material-increasing the speed of the procedure and reducing trauma to the patient since the guide catheter does not need to be reintroduced (e.g., advanced through the vasculature and past the heart) before each pass.

[0099] Moreover, in certain embodiments, the present technology can enable the guide catheter **206** to be relocated to an alternate treatment site within the patient without removing the guide catheter **206** from the patient and, therefore, without reintroducing the guide catheter **206** through the heart. For example, the guide catheter **206** can be relocated to another treatment site within the lungs including a treatment site in the opposite lung. More specifically, (i) a dilator can be reintroduced into the guide catheter **206**, (ii) the guide catheter **206** can be withdrawn into the main pulmonary artery, (iii) the guidewire can be redirected to the new treatment site, (iv) the guide catheter **206** can be advanced over the guidewire to the new treatment site, and (v) the dilator can be removed.

[0100] Additional details of the systems, devices, and methods described above with reference to FIGS. **11-14** are provided in Appendix C to this application.

V. EXAMPLES

[0101] 1. A method for the intravascular treatment of clot material from a treatment site within a blood vessel of a human patient, the method comprising: [0102] engaging a first interventional device with the clot material at the treatment site in the blood vessel; [0103] retracting the first interventional device and a portion of the clot material into a distal portion of an elongated shaft; [0104] inserting a first valve insert into an attachment member coupled to a proximal portion of the elongated shaft; [0105] withdrawing the first interventional device and the portion of the clot material proximally through (a) the elongated shaft, (b) the attachment member, and (c) the first

valve insert; [0106] inserting a second valve insert into the attachment member instead of the first valve insert; [0107] aspirating the elongated shaft; and [0108] advancing a second interventional device distally through the elongated shaft to the treatment site.

[0109] 2. The method of example 1 wherein, after inserting the first valve insert into the attachment member, the first valve insert and the attachment member together define a lumen having a generally constant diameter.

[0110] 3. The method of example 2 wherein the diameter of the lumen is generally the same as the diameter of the elongated shaft.

[0111] 4. The method of any one of examples 1-3 wherein inserting the first valve insert into the attachment member includes exercising a valve of the attachment member.

[0112] 5. The method of any one of examples 1-4, further comprising actuating the second valve insert to seal the attachment member.

[0113] 6. The method of any one of examples 1-5 wherein the first interventional device and the second interventional device are the same interventional device, and wherein the method further comprises cleaning the interventional device to remove the portion of the clot material.

[0114] 7. The method of any one of examples 1-6, further comprising engaging the second interventional device with remaining clot material at the treatment site in the blood vessel.

[0115] 8. The method of any one of examples 1-7 wherein aspirating the elongated shaft includes aspirating, into the elongated shaft, at least a portion of clot material remaining at the treatment site in the blood vessel.

[0116] 9. The method of example 8 wherein aspirating the elongated shaft includes— [0117] coupling a syringe to the elongated shaft via a fluid control unit, wherein opening of the fluid control unit fluidly connects the syringe to the elongated shaft, and wherein closing of the fluid control device fluidly disconnects the syringe from the elongated shaft; [0118] drawing a plunger of the syringe to generate a vacuum while the fluid control unit is closed; and [0119] opening the fluid control unit to apply the vacuum to the elongated shaft to thereby aspirate the portion of the remaining clot material into the elongated shaft.

[0120] 10. The method of example 9 wherein the fluid control unit is a clamp.

[0121] 11. A method for the intravascular treatment of clot material from a treatment site within a blood vessel of a human patient, the method comprising: [0122] engaging a first interventional device with the clot material at the treatment site in the blood vessel; [0123] retracting the first interventional device and a portion of the clot material into a distal portion of an elongated shaft; [0124] withdrawing the interventional device and the portion of the clot material from the blood vessel through the elongated shaft and an attachment member coupled thereto, including— [0125] before withdrawing the interventional device through the attachment member, actuating the attachment member to move the attachment member from a first configuration in which a lumen of the attachment member is sealed to a second configuration in which the lumen of the attachment member is open; [0126] after withdrawing the interventional device through the attachment member, actuating the attachment member to return the attachment member to the first configuration from the second configuration; [0127] aspirating the elongated shaft; and [0128] advancing a second interventional device distally through the elongated shaft to the treatment site. [0129] 12. The method of example 11 wherein actuating the attachment member includes pressing one or more buttons on the attachment member.

[0130] 13. The method of example 11 or 12 wherein in the second configuration, the lumen of the attachment member has a generally constant diameter.

[0131] 14. The method of example 13 wherein the diameter of the lumen is generally the same as a diameter of the elongated shaft.

[0132] 15. The method of any one of examples 11-14 wherein the first interventional device and the second interventional device are the same interventional device, and wherein the method further comprises cleaning the interventional device to remove the portion of the clot material.

[0133] 16. The method of any one of examples 11-15, further comprising engaging the second interventional device with remaining clot material in the blood vessel.

[0134] 17. The method of one of examples 11-16 wherein aspirating the elongated shaft includes—
[0135] coupling a syringe to the elongated shaft via a stopcock, wherein opening of the stopcock fluidly connects the syringe to the elongated shaft, and wherein closing of the stopcock fluidly disconnects the syringe from the elongated shaft; [0136] drawing a plunger of the syringe to generate a vacuum while the stopcock is closed; and [0137] opening the stopcock to apply the vacuum to the elongated shaft to thereby aspirate, into the elongated shaft, at least a portion of clot material remaining at the treatment site in the blood vessel.

[0138] 18. A method for the intravascular treatment of clot material from a treatment site within a blood vessel of a human patient, the method comprising: [0139] engaging an interventional device with the clot material at the treatment site in the blood vessel; [0140] retracting the interventional device and a portion of the clot material into a distal portion of a lumen of an elongated shaft; [0141] unsealing an attachment member coupled to a proximal portion of the elongated shaft; [0142] withdrawing the interventional device and the portion of the clot material through the lumen of the elongated shaft and through the attachment member; [0143] sealing the attachment member; [0144] aspirating the elongated shaft; and [0145] advancing the same or a different interventional device distally through the lumen of the elongated shaft to the treatment site.

[0146] 19. The method of example 18 wherein unsealing the attachment member includes exercising a valve of the attachment member such that a lumen of the attachment member has a diameter that is about equal to a diameter of the lumen of the elongated shaft.

[0147] 20. The method of example 18 or 19, further comprising: [0148] engaging the same or the different interventional device with clot material remaining in the blood vessel; and [0149] retracting the same or the different interventional device and a portion of the remaining clot material into the distal portion of the lumen of the elongated shaft; [0150] unsealing the attachment member; and [0151] withdrawing the same or the different interventional device and the portion of the remaining clot material through the lumen of the elongated shaft and through the attachment member.

VI. CONCLUSION

[0152] The above detailed descriptions of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology as those skilled in the relevant art will recognize. For example, although steps are presented in a given order, alternative embodiments may perform steps in a different order. The various embodiments described herein may also be combined to provide further embodiments.

[0153] From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments of the technology. Where the context permits, singular or plural terms may also include the plural or singular term, respectively.

[0154] Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. Additionally, the term “comprising” is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded. It will also be appreciated that specific embodiments have been described herein for purposes of illustration, but that various modifications may be made without deviating from the technology. Further, while advantages associated with some embodiments of the technology have been described in the context of those embodiments,

other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1-20. (canceled)

21. A method of aspirating clot material from a blood vessel of a patient, the method comprising: intravascularly positioning an aspiration catheter proximate to the clot material within the blood vessel; coupling a flow and aspiration assembly to the aspiration catheter, wherein the flow and aspiration assembly comprises— a reservoir a handle body; a pump assembly carried by the handle body, wherein the pump assembly includes a barrel and a plunger slidably positioned in the barrel; an actuation mechanism carried by the handle body and coupled to the pump assembly, wherein the actuation mechanism is actuatable to move the plunger in (a) a first direction through the barrel to generate negative pressure in the barrel and (b) a second direction through the barrel to generate positive pressure in the barrel; a first one-way valve configured to (a) inhibit fluid flow from the aspiration catheter to the barrel when the actuation assembly is actuated to move the plunger in the first direction and to (b) prevent fluid flow from the barrel to the aspiration catheter when the actuation assembly is actuated to move the plunger in the second direction; a second one-way valve configured to (a) inhibit fluid flow from the reservoir to the barrel when the actuation assembly is actuated to move the plunger in the first direction and to (b) permit fluid flow from the barrel to the reservoir when the actuation assembly is actuated to move the plunger in the second direction; and a filter assembly; and sequentially actuating the actuation mechanism to move the plunger in the first direction and the second direction to (a) aspirate blood and at least a portion of the clot material through the aspiration catheter, (b) drive the blood and the portion of the clot material into the filter assembly to filter the blood from the portion of the clot material, and (c) drive the filtered blood to the reservoir.

22. The method of claim 21 wherein the filter assembly comprises a housing and a filter removably positioned within the housing, and wherein the filter is configured to filter the blood from the portion of the clot material.

23. The method of claim 22 wherein the method further comprises removing the filter from the housing to access the portion of the clot material after sequentially actuating the actuation mechanism.

24. The method of claim 21 wherein the filter assembly comprises a housing and a filter positioned within the housing, wherein the filter is configured to filter the blood from the portion of the clot material, and wherein the filter has a cylindrical shape.

25. The method of claim 21 wherein the filter assembly comprises a first port fluidly coupling the filter assembly to the aspiration catheter and a second port fluidly coupling the filter assembly to the pump assembly.

26. The method of claim 21 wherein sequentially actuating the actuation mechanism to move the plunger in the first direction and the second direction pulls the blood and the portion of the clot material into the filter assembly via the negative pressure generated in the barrel.

27. The method of claim 21 wherein the actuation mechanism comprises a lever coupled to the handle body and to the plunger.

28. The method of claim 21 wherein the reservoir comprises a bag.

29. The method of claim 21 wherein the filter assembly comprises: a housing defining a sealed chamber and having a first housing end portion and a second housing end portion opposite the first housing end portion; a first port positioned proximate to the first housing end portion and configured to receive the blood and the portion of the clot material therethrough; a second port configured to receive the filtered blood therethrough; and a filter extending from a first filter end

portion to a second filter end portion, wherein the filter extends continuously about the first filter end portion such that the filter encloses an interior region around the second port, and wherein the filter is configured to inhibit the portion of the clot material from passing through the filter into the interior region and through the second port.

30. The method of claim 29 wherein the filter has a substantially cylindrical shape.

31. The method of claim 29 wherein the filter is removably positioned within the chamber, and wherein the method further comprises removing the filter from the chamber to access the portion of the clot material after sequentially actuating the actuation mechanism.

32. The method of claim 29 wherein the housing is at least partially transparent to permit visualization of the portion of clot material in the chamber outside the interior region.

33. The method of claim 29 wherein the second port is positioned proximate to the second housing end portion.

34. The method of claim 29 wherein the filter assembly further comprises a cap configured to be releasably coupled to the second housing end portion, wherein the cap is removable from the housing to provide access to the filter, and wherein the method further comprises removing the cap from the second housing end portion to access the portion of the clot material after sequentially actuating the actuation mechanism.

35. The method of claim 29 wherein the first filter end portion is curved.

36. The method of claim 21 wherein the flow and aspiration assembly further comprises a valve between the pump assembly and the aspiration catheter, wherein the valve is movable between (a) a closed position inhibiting fluid flow from the aspiration catheter to the pump assembly and (b) an open position permitting fluid flow from the aspiration catheter to the pump assembly, and wherein the method further comprises: moving the valve to the closed position before actuating the actuation mechanism to move the plunger in the first direction to store the negative pressure in the barrel; and moving the valve to the open position to apply the stored negative pressure to the aspiration catheter before actuating the actuation mechanism to move the plunger in the second direction.

37. The method of claim 21 wherein sequentially actuating the actuation mechanism to move the plunger in the first direction and the second direction comprises: actuating the actuation mechanism to move the plunger in the first direction; then actuating the actuation mechanism to move the plunger in the second direction; then actuating the actuation mechanism to again move the plunger in the first direction; and then actuating the actuation mechanism to again move the plunger in the second direction.

38. A method of aspirating clot material from a blood vessel of a patient, the method comprising: intravascularly positioning an aspiration catheter proximate to the clot material within the blood vessel; coupling a flow and aspiration assembly to the aspiration catheter, wherein the flow and aspiration assembly comprises— a filtered blood tube; a handle body; a pump assembly carried by the handle body, wherein the pump assembly includes a barrel and a plunger slidably positioned in the barrel; an actuation mechanism carried by the handle body and coupled to the pump assembly, wherein the actuation mechanism is actuatable by a user to move the plunger in (a) a first direction through the barrel to generate negative pressure in the barrel and (b) a second direction through the barrel to generate positive pressure in the barrel; a first one-way valve configured to (a) inhibit fluid flow from the aspiration catheter to the barrel when the actuation assembly is actuated to move the plunger in the first direction and to (b) prevent fluid flow from the barrel to the aspiration catheter when the actuation assembly is actuated to move the plunger in the second direction; a second one-way valve configured to (a) inhibit fluid flow from the filtered blood tube to the barrel when the actuation assembly is actuated to move the plunger in the first direction and to (b) permit fluid flow from the barrel to the filtered blood tube when the actuation assembly is actuated to move the plunger in the second direction; and a filter assembly; and sequentially actuating the actuation mechanism to move the plunger in the first direction and the second direction to (a)

aspirate blood and at least a portion of the clot material through the aspiration catheter, (b) drive the blood and the portion of the clot material into the filter assembly to filter the blood from the portion of the clot material, and (c) drive the filtered blood to the filtered blood tube.

39. The method of claim 38 wherein the filter assembly further comprises a cap configured to be releasably coupled to the second housing end portion, wherein the cap is removable from the housing to provide access to the filter, and wherein the method further comprises removing the cap from the second housing end portion to access the portion of the clot material after sequentially actuating the actuation mechanism.

40. The method of claim 38 wherein the filter assembly comprises: a housing defining a sealed chamber and having a first housing end portion and a second housing end portion opposite the first housing end portion; a first port positioned proximate to the first housing end portion and configured to receive the blood and the portion of the clot material therethrough; a second port configured to receive the filtered blood therethrough; and a filter extending from a first filter end portion to a second filter end portion, wherein the filter extends continuously about the first filter end portion such that the filter encloses an interior region around the second port, and wherein the filter is configured to inhibit the portion of the clot material from passing through the filter into the interior region and through the second port.
